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BRAIN- COMPUTER INTERFACES

U.S. MILITARY APPLICATIONS AND IMPLICATIONS

AN INITIAL ASSESSMENT

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Summary

Brain-computer interface (BCI) represents an emerging and potentially disruptive area of technology that, to date, has received minimal public discussion in the defense and national security policy communities. This research considered key areas in which future BCI technologies might be relevant for the warfighters of tomorrow. It sought to explore the operational value of current and future developments regarding man-machine neural communication, the associated vulnerabilities and risks, and the policy levers that should be in place before the technology is deployed.

The project drew from reviews of relevant technical and security literature and discussions with subject-matter experts to develop a July 2018 game convening technical and operational experts. The game tested the potential utility of a functional “BCI toolbox” against two future tactical urban operations vignettes. Game results indicated that BCI technologies are likely to have practical use on a future battlefield, particularly as the pace and volume of human-machine interaction intensify. Within the vignettes, participants anticipated that BCI capabilities could enhance the speed of communication, improve common situational awareness, and allow operators to control multiple technological platforms simultaneously. Participants noted that the pragmatic utility of each BCI capability would depend largely on its fidelity and reliability during combat. Of the capabilities assessed in the game, direct brain-to-brain communication facilitated by BCI appeared to offer the most transformative applications for operational use but also carried the most significant operational and institutional risks.

Our analysis also explored possible areas of risk associated with the development and application of BCI combat capabilities. As with many new technological developments, BCI may create new military operational vulnerabilities, new areas of ethical and legal risk, and potentially profound implications for existing military organizational structures. In particular, the report highlights potential operational vulnerabilities associated with the development and adoption of BCI technologies by the U.S. Department of Defense (DoD), including the potential for new

points of failure, adversary access to new information, and new areas of exposure to harm or avenues of influence of service members. It also underscores institutional vulnerabilities that may arise, including challenges surrounding a deficit of trust in BCI technologies, as well as the potential erosion of unit cohesion, unit leadership, and other critical interpersonal military relationships. Finally, we consider potential future U.S. government ethical and legal responsibilities to an individual BCI operator, as well as the implications that BCI technologies might have on the ethical and legal responsibilities of that individual. These considerations should be incorporated into research and development (R&D) efforts early in the process and may warrant dedicated a departmentwide oversight mechanism as the technologies continue to mature.

Overall, our findings suggest that as the U.S. military increasingly incorporates artificial intelligence (AI) and semiautonomous systems into its operations, BCI could offer an important means to expand and improve human-machine teaming. However, precautions will need to be taken to mitigate vulnerabilities to DoD operations and institutions and to reduce potential ethical and legal risks associated with DoD’s development and adoption of BCI technologies. Specifically, we recommend that DoD

- **expand analysis to illuminate operational relevance and risks.** This research developed a systematic approach to evaluating potential operational applications of BCI by pairing operational experience with technological expertise and incorporating a disruptive and creative Red team of RAND Corporation experts. New analytical approaches such as this could supplement existing internal exercises to help ensure that operational needs and risks, rather than just technical opportunities, drive BCI development and identify new adversary threats.
- **address the trust deficit.** The game and associated research highlighted the extent to which cultural barriers to BCI, particularly among infantry service members, are likely to be high. Trust barriers could be mitigated through heavy vetting and testing in

Abbreviations

AI	artificial intelligence
ARL	Army Research Laboratory
BCI	brain-computer interface
BMI	brain-machine interface
BRAIN	Brain Research through Advancing Innovative Neurotechnologies
CaN CTA	Cognition and Neuroergonomics Collaborative Technology Alliance
DARPA	Defense Advanced Research Projects Agency
DNI	direct neural interface
DoD	U.S. Department of Defense
EEG	electroencephalogram
EW	electronic warfare
FDA	U.S. Food and Drug Administration
HCI	human-computer interaction
IoT	Internet of Things
MMI	mind-machine interface
MOUT	<i>Military Operations in Urban Terrain</i>
N ³	Next-Generation Nonsurgical Neurotechnology
NCI	neural-control interface
NESD	Neural Engineering System Design
NIH	National Institutes of Health
OODA	observe, orient, decide, act
R&D	research and development
tDCS	transcranial direct current stimulation
TNT	Targeted Neuroplasticity Training
TTX	table-top exercise

noncombat scenarios, introduction to service members that already rely on machine technologies, and an initial focus on noninvasive measures and medical applications.

- **collaborate and anticipate.** Our research highlighted examples of where DoD seed funding yielded successful BCI breakthroughs, and examples of emerging private-sector innovation. Where possible, future collaboration could leverage private-sector advances to the benefit of the U.S. military and, if carefully pursued, could improve trust gaps within the military. As the commercial market develops BCI technologies, this will help establish its capabilities and shortcomings. Although BCI applications are currently still in the basic-research phase, development of other technologies by the military, including robotics, AI, and big data analysis, will need to consider the eventual availability of BCI.
- **plan ahead for institutional implications.** As the U.S. government prepares to incorporate BCI technologies into future military capabilities, it will require institutional innovations to address new ethical and policy issues at each stage of the process, from R&D to operational application to veteran care.

Introduction

The 86 billion neurons of the human brain represent humankind's primary evolutionary advantage and, perhaps, an area of untapped potential. Currently, our brains interact with the world through our bodies, sending electrical currents through the nervous system to vocalize with our mouths, to type—or swipe—with our fingers, or to move bipedally through space. What will happen when human brains are freed of their corporeal confines and can control machines directly? Neurotechnological advances have already given quadriplegics the ability to perform basic operations in an F-35 simulator with their thoughts¹ and have given scientists the ability to decode speech that subjects are imagining in their minds—albeit imperfectly. Eventually, our physical

bodies might become a constraint that could be circumvented with appropriate neurotechnology.² The technical means for this brain-body bypass are BCIs, defined as methods and systems for providing a direct communication pathway between an enhanced or wired brain and an external device, with bidirectional information flow (between the brain and a device).³ Their potential impact is broad and far reaching, and policies on how to develop and manage such technology should be proactive, not reactive.

BCI technology is progressing. Such progress highlights the need to assess current and potential applications, and to ensure that the technology responds to actual needs in addition to the intentions of developers. As BCI transitions from basic research to more operational and commercial applications, it will be important to devote early attention to the broader implications, to consider what policies and guidelines might maximize its benefits while mitigating potential downsides. Developing such technologies as AI, data analytics, and robotics have captured headlines and fostered public discussion regarding potential benefits and risks. Limited comparable conversation has, as of yet, evolved for BCI. When compared to other prominent emerging technologies, BCI is relatively immature; few capabilities have been deployed commercially. However, it has the potential to be no less influential. With profound potential implications in fields from defense and national security to health and wellness, BCI may represent a highly disruptive technology that, to date, has received insufficient analysis.

This report offers an initial assessment of what viable applications BCI may have in U.S. military operations, and what risks and vulnerabilities may be associated with its development and deployment. The authors describe the current state of the art and possible areas of technology development and growth for BCI military applications and investigate key questions associated with the use of BCI capabilities in a future combat scenario. Fundamentally, we ask, (1) what is the potential operational significance of current and future developments regarding BCI, and (2) what are the policy considerations necessary for effective management of the technology with an understanding of its potential impact on the warfighter of the future?

Human-Machine Teaming

The research views BCI in the context of the anticipated future of warfare, including increases in human-machine teaming. The analysis begins from the premise that human-machine teaming will play a major role in future combat and that BCI may provide a competitive advantage in future warfare. Former Deputy Secretary of Defense Robert Work, who led DoD's *3rd offset*, a catalyst for defense-sector technology development focused on human-machine teaming, summarized trends with military technology as follows:

The coin of the realm during the Cold War was armored brigades, mechanized infantry brigades, multiple launch rocket system battalions, self-propelled artillery battalions, tactical fighter squadrons, among others. Now, the coin of the realm is going to be learning machines and human-machine collaborations, which allows machines to allow humans to make better decisions; assisted human operations, which means bringing the power of the network to the individual; human-machine combat teaming; and the autonomous network.⁴

Although DoD R&D efforts include many dimensions of technology, certain aspects are particularly relevant to potential work with BCI. In particular, there is an increasing focus on human-machine collaboration for improved decisionmaking, including human-computer interaction (HCI) and cognitive teaming, assisted-human operations, and advanced manned and unmanned combat teaming.⁵ Defense officials discussing the “centaur” model—collaborative human and AI teams—have highlighted the relative advantages of the U.S. civilian and military workforce in developing and operating human-machine teaming technologies.⁶ Human-machine teaming technologies that effectively leverage the unique cultural strengths of the American warfighter, including critical thinking and creative problem-solving, represent an area of particular value to the future U.S. military.⁷ Technology development programs within DoD have thus focused on human-machine collaboration.⁸ In fact, the Defense Advanced Research Projects Agency

Ultimately, humans and machines could collaborate cognitively and seamlessly—to think together.

(DARPA) has suggested that “smart systems will significantly impact how our troops operate in the future, and now is the time to be thinking about what human-machine teaming will actually look like and how it might be accomplished. . . .”⁹

Future developments with human-machine teaming have the potential to prompt far-reaching defense policy debates.¹⁰ Technological advances, such as BCI, that allow humans to connect increasingly closely with machines on the battlefield may yield fundamental strategic and operational changes within each of the U.S. service branches and will undoubtedly raise ethical and organizational questions across the U.S. defense community. Thus, as DoD pursues a forward-looking vision for human-machine teaming, ambitious planning should prompt similarly far-reaching defense policy debates.¹¹

In preparation for a future world in which human-machine teams represent the “coin of the realm,” DoD has already invested in the development of technologies that can permit the human brain to communicate directly with machines, including the development of implantable neural interfaces that are capable of transferring data between the human brain and the digital world.¹² On the future battlefield, human thoughts may well be channeled to AI software or to robots, with information transferred back from sensors and machines directly to the human brain.¹³ Ultimately, humans and machines could collaborate cognitively and seamlessly—to think together.

Approach

This analysis is structured to explore the operational implications of BCI technology. It pilots a repeatable process for systematically exploring the relevance and implications of emerging technologies in the context of military operations. Mapping technological capabilities to practical applications—and understanding not just the state of the art but the state of the practical uses—can present a significant challenge for assessments of emerging technology. Our process addresses that challenge by parsing the technology into operationally relevant capabilities, testing their operational relevance with a table-top exercise (TTX), and then exploring implications, risks, and risk-mitigation strategies. The process is itemized as follows:

1. Through literature review and discussions with subject-matter experts, we summarize the technology and identify key areas of developmental effort.
2. Specific topics of technology development are used as a catalyst for discussions with military experts to identify potential military applications in theater.
3. We aggregate the results of the analysis concerning technology development and operational relevance to provide a general assessment of whether BCI could potentially be valuable in a military setting and, if so, how. This preliminary finding becomes the overarching guide for a TTX.
4. Based on the technical and operational assessments, we derive a set of projected BCI capabilities—a future BCI toolbox.
5. The BCI toolbox is used in a table-top game based on exemplary scenarios involving tactical urban operations. This game explores more extensively the anticipated viability and relevance of BCI in theater.
6. Throughout this process, we explore vulnerabilities, risks, and risk-mitigation strategies associated with BCIs across three dimensions: technological, institutional, and legal/ethical.

Technology Summary

Introduction

Although a comprehensive review of the field of BCI would be beyond the report's scope, and although the literature on the topic is extensive enough to support handbooks,¹⁴ dedicated societies,¹⁵ and dedicated journals,¹⁶ this section highlights prominent work, themes, and organizations to inform the assessment of potential military applications. By drawing on technical and popular literature, as well as discussions with subject-matter experts, we segment the topic into distinct areas of work, clarifying what exactly BCI entails and what kind of research is active. The section ends with a discussion of future trends and potential directions and summarizes primary technical challenges and risks.

While we adopt the term *BCI* based on its prevalence in the literature, a variety of related terms are used to describe similar capabilities: neural-control interface (NCI), mind-machine interface (MMI), direct neural interface (DNI), and brain-machine interface (BMI).

Although there is some debate regarding the precise nature of signals that are transmitted within the human brain, BCIs generally involve monitoring or affecting those signals.¹⁷ Different BCI tools allow users to access and to use these signals with various levels of accuracy and invasiveness. In short, a BCI enables a bidirectional communication between a brain and an external device, and there is a broad range of ongoing work on this topic. In this context, bidirectional generally includes direct neural readout and feedback and direct neural write-in.

As reflected by the aforementioned definition and by the alternate terms, the focus is often on human-machine teaming, which aligns with needs stemming from trends in warfare. Practically, BCI provides a mechanism for blending human strengths and computer strengths, and much of the ongoing work strives to link these two sets of capabilities and yield synergistic advantages. The efficiency of the interface between humans and machines—whether facilitating communication by screens, text, or another form—is a significant factor in allowing

humans to manage increasingly complex systems and information. BCI can improve such efficiency.

Although human-machine teaming can be useful, it is just a subset of BCI applications. The literature regarding ongoing R&D and potential applications extends beyond human-machine teaming, and BCIs need not just link humans and machine to provide value. More generally, BCIs provide a method to connect to the human brain. They provide more data. This connection can then be tied to a machine, to software, to another human, or simply to an output system for assessment. In fact, while human-machine teaming remains a cornerstone of developing technology for warfare, the broader advantages of BCIs point toward not just integrating humans and machines but leveraging human capabilities in general.

Review

While the practical significance of BCI has just recently become more visible, work in this field has been ongoing for nearly a century. In fact, work with the first human electroencephalogram (EEG)—a device for tracking and recording brain wave patterns—was published in 1929.¹⁸ Jacques Vidal coined the term *brain-computer interface* in 1973, and research in this field has continued since.¹⁹

Work with BCI tends to fall into the following categories, which provide a framework for our investigations of operational relevance and applied capabilities in subsequent sections:

- data transfer *from* the brain
- direct system control
- prosthetics and paralysis treatment
- cortically coupled AI (for training or running AI systems)
- data transfer *to* the brain, and brain-to-brain communication.

Each of these topics can be segmented further into work involving invasive systems and noninvasive systems. Invasive systems involve implanting electronic devices beneath the human skull, inside the brain. The surgery allows practitioners to place the implant exactly where desired to monitor precise sets of neurons that govern specific neurological functions, but it

carries health risks. Alternatively, noninvasive systems sit outside the skull. While this reduces risk to the user, the skull essentially acts as a filter and muffles the electrical signal.²⁰ The signals picked up by external electrodes are less clear, and it is more difficult to ascertain which neurons are firing.

A key effort in fueling most of these topics has been the Brain Research through Advancing Innovative Neurotechnologies (BRAIN) initiative at the National Institutes of Health (NIH). This effort is broadly aimed at “revolutionizing our understanding of the human brain.”²¹ BRAIN partners include the National Science Foundation, DARPA, the U.S. Food and Drug Administration (FDA), and the Intelligence Advanced Research Projects Agency (IARPA), as well as foundations, institutes, universities, and industries. NIH allocated \$46 million in 2014 and \$81.4 million in 2015, reflecting growing interest in the topic.²² Overall, DARPA has invested “hundreds of millions of dollars” transitioning *neuroscience* into *neurotechnology* since the early 2000s.²³

With regard to the topic of *transferring data from the human*, a primary goal is assessing cognitive performance. To this end, the Army Research Laboratory (ARL) is using 3-D printing to create helmets that fit perfectly to each user and then incorporate EEG sensors to monitor brain activity.²⁴ The Air Force is also pursuing a camera-based comprehensive cognitive monitoring system built into a pilot’s helmet to monitor cognitive workload and stress.²⁵ The helmets can adapt displays based on the pilot’s unique physical and mental conditions. Separate research sponsored by ARL has investigated deep learning solutions for predicting drowsy and alert states based on EEG readings, and DARPA-sponsored teams have tested “closed-loop” brain implants that use algorithms to detect patterns associated with mood disorders.²⁶ BCI devices to monitor performance and even emotional spikes related to depression, anxiety, or rage are increasingly prevalent in China, where uses include factories, public transport, state-owned companies, and the military.²⁷

Related to systems for extracting data from the human brain for assessment are methods for *direct system control*, whereby users control machines wirelessly with brain activity. In one well-publicized

example, DARPA, APL, and the University of Pittsburgh used a BCI implant to permit a quadriplegic woman to operate flight simulators.²⁸ Researchers have also corrected robot mistakes through noninvasive measurement of EEG signals.²⁹

Other research projects focus on drone control. With funding from DARPA and the U.S. Army, researchers at the Human-Oriented Robotics and Control laboratory enable a user to control a swarm of drones.³⁰ The lab’s researchers suggest the technology could be used practically in the military within five to ten years. Applications also include delivery of medical help, search and rescue, and exploration, all in remote or inaccessible environments. Finally, using commercial hardware, researchers at the University of Florida have constructed, and demonstrated the use of, a low-cost system that is capable of wirelessly controlling common small drones.³¹

Beyond military applications, the health-care sector has advanced significant work with BCI direct control, especially involving invasive systems. Work at Stanford University enables paraplegic patients to control a computer mouse and computer software with their thoughts.³² Dr. Krishna Shenoy, one of the principal investigators, suggests, “The day will come—closer to five than 10 years from now—when a self-calibrating, fully implanted wireless system can be used without caregiver assistance, has no cosmetic impact and can be used around the clock.”³³

The integration of *prosthetics*, which is essentially a subset of the work with direct systems control, has direct applications in the health-care sector. Much of this work involves invasive systems, primarily because of the necessity to target specific sets of neurons with relatively high accuracy. Perhaps the most complex prosthetic, conceptually, is the spinal cord. Researchers have used electrodes to reconnect the motor cortex and the spinal cord in monkeys and rats, restoring the ability to walk.³⁴ Case Western Reserve University used a similar procedure, called *functional electrical stimulation*, to enable arm and hand movement by bypassing the spinal cord and directly inducing muscle stimulation.³⁵ As part of the Revolutionizing Prosthetics program, DARPA and APL have pursued research concerning BCI-enabled prosthetic hands and arms that use intracortical microstimulation to provide feedback to the user (directly to the brain) and

evoke sensations that the user perceived as coming from his/her own hand.³⁶ APL extended this work to study how BCI users could interpret external artificially provided intracortical microstimulation, even if the type of information being provided varied from what a brain region would normally have been processing. They then used this approach to provide a BCI user with navigation information from a simulated Mooney Bravo aircraft.³⁷

Data (or information) from a human brain can be used not only to inform assessment tools or to drive systems but also to inform software with *cortically coupled AI*. Rather than using brain signals to *control* a computer or a system, a “cortically coupled computer system opportunistically senses the brain state, capturing a user’s implicit or explicit computation, and then communicates this information to a traditional computer system via a neural interface.”³⁸ This information can then potentially help train an AI system. This use of BCI represents a heightened level of human-machine teaming, allowing a human to *think with* a machine (or a computer) or, more specifically, integrate human thoughts or data into a process conducted by machine. Real-time BCI interaction could negate the current requirement for predetermined computer codes to transfer information, addressing one of the key bottlenecks of traditional human-machine integration.³⁹ Such teaming is of particular interest to members of the AI community who are exploring methods and approaches for managing and “controlling” AI. BCI may be able to provide this tool. Elon Musk, the founder of Neuralink, a relatively new company focused on integrating humans with AI, suggests, “Some high bandwidth interface to the brain will be something that helps achieve a symbiosis between human and machine intelligence and maybe solves the control problem and the usefulness problem. . . .”⁴⁰

In addition to the extraction of data from the brain, there is also work exploring the ability to implant or *transfer information to the brain* itself. A significant challenge in enabling efficient system and prosthetic control is feedback to the user, providing information about the system being controlled. Although patients in a lab setting may be able to control a prosthetic hand using a BCI, for example, they will not necessarily realize where their

Noninvasive transcranial direct currents can be used to treat depression and strokes, to increase focus and attention, to shorten training time, and, potentially, to improve physical training.

prosthetic hand is without seeing it, unless they can look directly at it to obtain visual feedback.

Research at Wake Forest Baptist Medical Center and the University of Southern California, funded by DARPA’s Restoring Active Memory program, has shown initial success improving memory using surgically implanted electrodes, a strand of research with promise for treating Alzheimer’s disease, strokes, and head injuries.⁴¹ By reinforcing recorded neural patterns from a patient’s experience (i.e., seeing a particular image), researchers were able to improve episodic memory, which is the most common type of memory loss in people with Alzheimer’s disease, stroke, and head injury. Researchers saw a 35-percent improvement in subjects’ baseline short-term memory.⁴² Researchers noted, “This is the first time scientists have been able to identify a patient’s own brain cell code or pattern for memory and, in essence, ‘write in’ that code to make existing memory work better, an important first step in potentially restoring memory loss. . . .” While this work focused on improving existing memory skills, future work might enhance the ability to retain specific memories as memory skill begins to fail. Noninvasive transcranial direct currents can be used to treat depression and strokes, to increase focus and attention, to shorten training time, and, potentially, to improve physical training (with a focus on the motor cortex).⁴³

A natural extension from research that aims to read brain signals and to send or implant information in the brain is brain-to-brain communication.

The DARPA Neural Engineering System Design (NESD) program is developing invasive systems that can communicate clearly and individually with any of up to one million neurons in a given region of the brain, and this includes the ability both to transmit to the brain and read from the brain with some neurons.⁴⁴ While current invasive devices may incorporate something on the order of 100 channels, this project strives to read 10^6 neurons, write to 10^5 neurons, and interact with 10^3 neurons full-duplex, a far greater scale than is possible with existing neurotechnology.⁴⁵ Another DARPA program, the Next-Generation Nonsurgical Neurotechnology (N³) program involves a noninvasive system capable of reading from and writing to multiple points in the brain at once.⁴⁶

A natural extension from research that aims to read brain signals and to send or implant information in the brain is *brain-to-brain communication*. With funding from the ARL, researchers at the University of Washington conducted a pilot study for a noninvasive system that uses EEG to read basic brain signals, transmit them over the internet, and then transfer motor responses to a second user using transcranial magnetic stimulation.⁴⁷ The signals represent very basic actions in the context of a simple video game, such as *move left* or *right*. Nonetheless, especially given that these signals are transferred over the internet, the potential to send even basic thoughts across the internet inherently presents many opportunities and many risks concerning security and ethics. The

authors extend this effort to involve five groups of three individuals.⁴⁸ Two individuals sent information, and the third person received information, with all three collectively playing a Tetris-like game. The work follows earlier experiments in transferring signals between rats⁴⁹ and from a human to a rat.⁵⁰

Development Directions and Technical Challenges

Development Directions

In general, the direction with most BCI work concerns the amount and quality of data being transferred. The fidelity with which data can be extracted and transferred from the human brain will likely increase.⁵¹ Signal bandwidth will likely improve. DARPA's NESD program, for example, has invested in research on implantable neural interfaces for sensory restoration that may engage up to one million neurons at once.⁵² As of July 2019, one of the grant recipients, led by Brown University, presented 0.25-square-millimeter implants, called "neurograins," that permit wireless bidirectional communication with an external device with an uplink rate of up to 10 megabits per second.⁵³ As of January 2020, another grantee, Paradromics, had announced a new high-data rate implantable BCI that can process and transmit neural data at 60-times-lower power dissipation than existing approaches, allowing transfer of more data at lower risk of overheating the brain.⁵⁴

Although no proof-of-concept currently exists for this technology, one potential frontier in transferring data from the human brain may be long-distance standoff wireless assessments. These might allow commanders to assess their soldiers' state or even the enemy's state from long distances. Similarly, BCI might find use in aggregating the assessment of a group. For example, BCI could be used to monitor the cognitive workload of a squad. For direct system control, more work will be needed to transfer complex manipulations or strategies with resistance to distractions.

With respect to prosthetics, the next step is establishing new neurological connections. Currently, BCI efforts regarding prosthetics involve

reconnecting existing neurons to physical systems. It is more challenging to provide the ability to control a prosthetic (and associated neurons) that never existed. In addition, more work is also needed to provide proprioceptive feedback directly to the brain to improve bidirectional BCI control of prosthetic limbs. For data transfer to the brain, the ultimate goal is to provide the brain with direct, high-fidelity information (e.g., isolated memory implants). Like long-distance standoff assessments, this capability is not yet feasible but nonetheless remains a target for the BCI field.

A particularly interesting prospect for BCI is its integration with the Internet of Things (IoT), which connects systems via the internet. DoD believes IoT can contribute to improved readiness by allowing one to monitor the status of materiel and weapons systems in real time, and it is thus becoming pervasive.⁵⁵ IoT has tactical applications, including giving warfighters access to sensors and data, and BCI could enhance this ability.⁵⁶

Some argue that, ultimately, the direction of this technology will follow the market.⁵⁷ With the increasing focus on entrepreneurship, researchers and academicians may likely spin off new technology, and commercial entities will drive development based on market demand. To be sure, companies like Kernel,⁵⁸ Neuralink,⁵⁹ Paradromics,⁶⁰ and Facebook⁶¹ are actively pursuing BCI capabilities. A scenario whereby commercial industry dominates this space can have two modes. Industry can, of course, respond to needs that pull the technology, but industry may also push technology according to anticipated profitability. It may be advantageous to have the relevant policy in place *before* the marketplace drives the technology.

Technical Challenges and Risks

Despite the exciting and dynamic future that BCIs may uncover, there are, of course, technological challenges and risks. Perhaps the most significant technological challenge in BCI development is the trade-off between signal clarity and the ability to target specific neurons provided by invasive systems, and the ease of use with noninvasive systems.⁶² DARPA's N³ program is currently seeking to address some of

these challenges by developing a portable noninvasive system capable of reading from and writing to multiple points in the brain at once.⁶³

Invasive systems, which provide higher-fidelity signals, carry risks associated with any surgery, including hemorrhaging, infection, or brain damage. Electrodes can also induce infections and can degrade with time. Scarring and exhaustion (when neural substrates stop reacting) reduce signal strength. Biocompatibility is also a significant limitation. Furthermore, all current implants corrode and, thus, limit the duration for which they are useful. Sensors for most implanted BCIs currently last only about two to five years, although some primate work involves sensors receiving signals for as long as seven to eight years. Reducing the sensor rigidity, size, and tendency for deterioration while retaining signal quality remains a persistent challenge. Developing sensors with additional channels to improve accuracy and minimizing sensor power usage, which can cause tissue damage, are additional challenges. Precise placement of sensors on the brain is also challenging. In general, the hardware necessary for BCI (amplifiers, cables, sensors, etc.) is still too large for practical use outside a lab.

With both invasive and noninvasive systems, the data gathered from neurons are then analyzed, and accurate decoding presents another challenge with the advancement of BCI. This decoding often involves some form of machine learning for performance assessment, which breaks down if the individual, task, or time frame changes. Furthermore, decoding algorithms are unstable and require regular recalibration, in part because the position of neurons relative to electrodes changes, and firing patterns naturally change. To be sure, machine learning is an active area of research, including methods for generalizing results such that an algorithm trained on one set of data may be used with acceptable accuracy on a new, slightly different set of input data.

Despite the aforementioned technical risks that must be addressed, the applications and underlying capabilities currently being explored with BCI suggest that BCI *may* have viable uses for warfare. Subsequent sections thus consider operational repercussions of using BCIs in a military setting.

Operational Considerations

Introduction

Military theorists and practitioners have long lauded the human mind as a critical determinant of military success. Could direct linkage between the brain and an external device improve a warfighter's performance? Building on the overview of emergent BCI technologies in the previous section, the following section begins to explore how BCI technologies might be applied within a military context.

Two lenses frame the discussion. The first considers relevant features of the future warfighter experience to extrapolate potential applications for BCI. The second draws from open-source material from U.S. military research organizations and discussions with subject-matter experts to identify existing concepts for how BCI might be applied even in today's operational environment. This analysis informs the introduction of a BCI toolbox, which serves as the conceptual cornerstone for the RAND Corporation's BCI TTX.

BCI and the Future Warfighter

In his work on the history of the future of war, British scholar Sir Lawrence Freedman notes that "there is no longer a dominant model for future war, but instead a blurred concept and a range of speculative possibilities."⁶⁴ His work does identify a few recent strategic themes addressed in the literature on future wars, including an increased prevalence of hybrid wars, cyberwar, use of robots and drones, and the advent of megacities and climate change as sites and catalysts for future wars.⁶⁵ A 2019 RAND Perspective further projects several broad strategic trends for the future of war: increased competition for regional hegemony, difficulty defending isolated countries and domains, a decline of American qualitative and quantitative military edge, blurred lines between war and peace, and continued war on terrorism.⁶⁶ Additionally, increased focus on inter-state strategic competition and the potential for conflicts with near-peer adversaries also raises the possibility that future adversaries may wield more resources and sophisticated technological capabilities than have

those of recent decades. However, as a 2019 Royal United Services Institute report on the future operational environment concludes, counter-Western strategies are "dynamic and evolving" and "differ depending on the context, means, ways, and ground."⁶⁷

At the operational level, the character of warfare and combat experiences of the human warfighter may be driven in large part by rapid advances in military technological innovation.⁶⁸ As both the United States and potential adversaries develop and deploy new battlefield technologies, collaborative relationships between humans and machines are likely to evolve and place new requirements on the cognitive workload for a future warfighter. Regarding the potential application of BCI, the future warfighter is likely to have increased requirements to

- digest and synthesize large amounts of data from an extensive network of humans and machines
- make decisions more rapidly due to advances in AI, enhanced connectivity, and autonomous weaponry
- oversee a greater number and types of robotics, including swarms.

One recurring theme in discussions of cognition and future combat is that decisionmaking is likely to be complicated by the synthesis of enormous amounts of information.⁶⁹ During ground combat, for example, information sources might include such tools as acoustic sensor networks, which could provide the location of distant gunfire, or drone trackers to detect swarming robots.⁷⁰ In a future battlefield defined by the IoT, smart devices, soldier-worn sensors, and unmanned aircraft may flood service members with actionable data.⁷¹ Extensive data and new sources of information may improve future situational awareness but could also complicate considerations for operational decisionmakers to process. Enhanced connectivity will expand the sources and speed of information transferred between humans or between humans and machines. Such information could range from networked connections between an F-35 and fourth-generation counterparts during an operation to real-time readiness status updates of ground-based materiel and weapon systems.⁷²

Unsurprisingly, services are already pursuing ways to facilitate rapid and extensive flow data between warfighters and decisionmakers to improve the smooth functioning of interconnected military systems.⁷³ BCI systems could serve as a potential future tool support this endeavor, allowing human analysts and operator to monitor and exploit larger amounts of information more effectively.⁷⁴

To address the potential for information overload, future service members are likely to engage more extensively with AI. On the future battlefield, AI tools may help human operators assess an environment, curate data, and ultimately allow operators to digest greater volumes of information.⁷⁵

Already, the U.S. Army has sought to leverage AI to “lighten the cognitive load” on future warfighters as a core capability objective of its 2017 Robotic and Autonomous Systems strategy.⁷⁶

As AI and rapid connectivity are increasingly incorporated into military operations, the pace of warfare will continue to accelerate.⁷⁷ Thus the speed at which decisions need to be made will also accelerate. In the coming decades, the United States and near-peer competitors are likely to seek out new ways to speed up decision cycles.⁷⁸

Finally, the future human warfighter may need to oversee and interact with a larger number of autonomous and semiautonomous systems.⁷⁹ Drone swarms may be incorporated at the tactical and operational levels in complex urban environments.⁸⁰ Future ground operations will incorporate robotics into supply and logistics chains.⁸¹ Already, service members can look to uninhabited aircraft flying above the combat zone for ISR and close air support. Additional combat applications for machines could include, for example, a robot that would be the first to enter a building and take fire, currently one of the deadliest roles in urban warfare.

Potential BCI Applications in Future Combat

Within the context of these technological and operational trends for military environments, this section summarizes potential relevant applications, based on the literature and feedback from subject-matter

To address the potential for information overload, future service members are likely to engage more extensively with AI,

experts. In general, BCI could theoretically be applied to help future warfighters make more informed decisions within a shorter timetable or to more effectively engage with more robotic systems than their current counterparts.

Laboratory studies indicate that BCIs may be able to enhance both the speed and accuracy of human decisionmaking.⁸² In a future BCI team, AI could theoretically transfer initial data analysis from a plane or drone directly to the relevant centers of an operator’s brain to further reduce cognitive load.⁸³ In combat, BCI could thus accelerate an operator’s observe, orient, decide, act (OODA) loop, through new ways of presenting information and bypassing physical senses.⁸⁴ Thus, DARPA cites the potential ability of military personnel to “facilitate multitasking at the speed of thought” and “interface with smart decision aids” as two rationales for its investment in noninvasive or minimally invasive BCI technologies.⁸⁵

BCI could also be used for more efficient engagement with AI to help maintain human oversight over operational decisions within a compressed time frame. Some scholars have hypothesized that an AI-enabled battlefield could lead to a phase shift in warfare in which the tempo of operations outpaces the speed of human decisionmaking. Some Chinese scholars have referred to this as a *battlefield singularity*.⁸⁶ Some U.S. scholars have referred to this concept as *hyperwar*.⁸⁷

If this hypothesis about the role of AI and automation in warfare is accurate, it may be the case that BCI is the only way to have humans remain

effectively engaged in decisionmaking in war and keep pace with machines. In this world, adopting BCI and effectively integrating humans with machines is not merely a tactical advantage but the central strategic advantage in warfare. BCI systems could facilitate *centaur* warfighting, leveraging “the precision and reliability of automation without sacrificing the robustness and flexibility of human intelligence.”⁸⁸ On the battlefield, one critical question would be whether BCI would permit humans to make meaningful decisions within future AI-driven operations tempos.

Additionally, BCI could yield potential advantages for human operators seeking to manage future robotics machines, or groups of machines, in combat. As former DARPA program manager Al Emondi has suggested, “As we approach a future in which increasingly autonomous systems will play a greater role in military operations, neural interface technology can help warfighters build a more intuitive interaction with these systems.”⁸⁹ As a practical matter, the ability to achieve hands-free control of a vehicle, robot, or a drone swarm through BCI could allow operators to use their hands for other tasks, such as carrying a traditional weapon. BCI could also potentially allow operators to do more with a swarm than manual operation would permit. One 2009 North Atlantic Treaty Organization study concluded that the goal of having a single human operator control multiple vehicles was “at best, very ambitious, and, at worst, improbable to achieve.”⁹⁰ Current work on brain-swarm interface posits that brain-computer technologies may be able to improve this challenge.⁹¹

Existing Concepts for BCI Combat Applications

Even today, BCI technologies, if they were available and more readily deployed, could yield specific operational benefits because of the direct access they permit to the human brain. Some DoD research programs publicly identify potential military operational applications of BCI that could be relevant in the existing operational environment. The following section draws from these themes, as well as insights

from subject-matter experts, to discuss additional potential applications for BCI during combat

One area where BCI technology could potentially prove useful for today’s military personnel would be *synthetic telepathy* among human operators.⁹² In 2009, DARPA’s “Silent Talk” program awarded grants to research institutions to “allow user-to-user communication on the battlefield without the use of vocalized speech through analysis of neural signals,” an application that could greatly facilitate covert communication.⁹³ An external analysis highlights the potential use of BCI technology to develop shared consciousness within and across units, improve collective awareness of combat challenges, and provide combatants with insights into perspectives and internal deliberations of multiple operators.⁹⁴

Direct access to the human brain could also help commanders improve the understanding of the cognitive and psychological states of their forces. As early as 2008, the Air Force investigated battlefield command-and-control systems that used EEG and eye movements to “assess the operator’s actual cognitive state” in an effort to “avoid cognitive bottlenecks before they occur” and eventually to “anticipate future mission state and operator functional state ahead of time.”⁹⁵ In its vision statement, the ARL’s Cognition and Neuroergonomics Collaborative Technology Alliance (CaN CTA) makes the case for developing the capacity to continually monitor operator neurocognitive behavior, including depth, distribution, and shifting of human attention, appraisal of information, the emotional context of actions, and the impact of physiological state—fatigue, stress, arousal—on cognitive and motor performance.⁹⁶ This type of function could plausibly identify and facilitate operations for extremely fatigued convoy drivers, or perhaps for gunners or tankers operating in complex environments for whom mistakes could prove deadly.⁹⁷ At a more complex level, a technology that could provide insights into the emotional state of a soldier might provide red flags as to whether and when the soldier might “break” psychologically, when a soldier might have psychotic tendencies, or perhaps when a soldier is shooting to miss. One study on the use of functional magnetic resonance imaging to identify falsehoods suggests that the ability to detect whether a subject is concealing information